Conduction by Hopping in RealQM:
the Free Boundary as Carrier

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August 27, 2026

Mathematical formulation by the author; analysis developed with AI assistance (Claude, Anthropic).

Abstract

RealQM models matter as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy. This article asks whether such a construction **conducts**.

What moves is not an electron. An electron here is a unit of charge density on a domain; what exists is the *partition* of space, and the carrier is a **free boundary** — in the relation a dislocation bears to the slip it produces, charge arriving at the far end while nothing material crosses the sample. Single occupancy makes the construction the Hubbard model at $U \to \infty$, so metals are out of scope in kind and not by degree, their carriers being extended Bloch states; doped Mott insulators, molecular semiconductors and polaronic oxides are not.

The activation energy for that boundary to advance one site can be computed, and it decides the question. Two configurations of a doped chain are fixed by *symmetry* — the carrier centred on a site, and centred between two — so both are stationary points at which the interface condition holds without being imposed, and their difference is the hopping barrier. The path itself carries a free parameter — the width over which the slip is spread — so the barrier is the minimum over that family and not a value at one width: $E_a = 555$ meV $= 21\, k_BT$, at a kink width of about one and a half sites, independent of chain length to 1.5%. The width dependence has a genuine minimum, elastic stiffness penalising a wide kink and the lattice's grip a narrow one, and every figure computed at a fixed width is an upper bound accordingly. The barrier *rises* under compression, to 1723 meV at two-thirds the spacing, so no metallic state appears at higher density. The carrier sits between sites rather than on one. At 24 to 67 k_BT across the range tested, against the 10–500 meV of real hopping semiconductors, the construction describes a **Mott insulator** and not the semiconducting class this article set out to claim.

A second obstacle is computational, and it conditions everything else. A free boundary evolves by *local* rules and so descends to the nearest configuration no single cell change improves — a stuck point, not a minimum — and which one depends on where the run started. Four independent geometries spread 0.005–0.05 Ha across seeds and builds, against the 0.001–0.02 Ha barriers sought; the ambiguity exceeds the quantity by one to two orders, and is unrepaired by surface tension, by step size, by freezing the labels, and by changing system. It is a failure of the *search*, not of the energy functional. The repair is to specify the partition rather than find it: place the domain walls, freeze them, relax only the wavefunctions. Repeated runs then agree to all five decimals recorded. On that footing a misplaced domain wall costs 179 meV, independent of chain length to 5%; a chain carrying a vacancy gains 0.112 Ha by delocalising its mismatch, also length-independent; and the barrier to sliding a pattern through its lattice is

tunable through zero by density and by core radius. The activation energy for a kink to *move*, which is what governs conductivity, is not among them and is left open with a stated route rather than estimated.

Two further results bound the framework. The interface coefficient vanishes exactly, $C = 0$, removing degeneracy pressure and leaving a vanishing screening length and a dispersionless plasmon; a Robin condition at $\beta a = 1.146$ restores $C = C_{\mathrm{TF}}$, but β that large *unbinds* H_2, which fails by $\beta a \approx 0.71$ — so the equation of state and molecular binding appear not to be repairable together, while double occupancy, the repair conduction needs at half filling, becomes favourable at $\beta a = 0.451$. And against density-functional theory and exact full configuration interaction on the same chains, the framework errs by **over-localising** — in bond length and in pinning alike — which is the direction $U \to \infty$ implies, so it fails as its own analysis says it should rather than arbitrarily.

1 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system, and the question pursued here is the elementary one such a system poses — whether it *conducts*, and if so what carries the charge. The same construction covers a stellar interior and the recombination era of a Coulomb cosmology, which are treated as the classical plasma, where $\omega_p^2 = 4\pi n$ comes out exactly.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla \psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|} \psi_i^2 \Big) dx + \tfrac{1}{2} \sum_{i \neq j} \iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x - y|} \, dx \, dy, \tag{1}$$

subject to $\int_{\Omega_i} \psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

A two-component system of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 What kind of conductor this can be

The construction does not have a free choice of target, and identifying it correctly settles most of what follows. One electron per domain, non-overlapping, is usually described as the Hubbard model at $U \to \infty$: double occupancy forbidden, every site singly occupied. That is right as far as it goes and it is not far enough. The Hubbard Hamiltonian has two parameters, and non-overlap fixes both. $U = \infty$ because two electrons may not share a domain; but also $t = 0$, because the transfer integral *is* the overlap of neighbouring orbitals and there is none. The construction is therefore not the Mott limit but the **atomic limit**, and the difference is not a refinement:

	$U \to \infty$, $t \neq 0$	$U \to \infty$, $t = 0$
half filling	Mott insulator	isolated atoms
superexchange	$J = 4t^2/U$	none
doped	**conducts**	still nothing
spin dynamics	yes	none

A Mott insulator is insulating but alive: it carries magnetic exchange, and doping it produces a conductor, which is what the cuprates are. The atomic limit is a collection of atoms with electrostatics between them. This accounts for every negative result below, and for one in particular: doping a Mott insulator works *through* t, so at $t = 0$ there is nothing for doping to activate — which is exactly what the measurement shows, the barrier rising with carrier concentration rather than falling. Metals are therefore out of scope in kind and not by degree. Their carriers are extended Bloch states, their transport is band motion, and their defining measurements — de Haas–van Alphen oscillations, a linear electronic specific heat, positive plasmon dispersion — belong to a degenerate electron gas with a sharp Fermi surface, which no arrangement of densities on domains appears able to supply. Two results usually read as failures follow from the scope rather than against it: $d\sigma/dT > 0$ is *correct* for an activated conductor and fatal only for a metal, and the absence of a Fermi surface does not bear on a dilute carrier gas, which is non-degenerate and has none to reproduce.

What remains within reach is matter whose carrier is a localised charge on a site, whose gap is on-site repulsion rather than band structure, and whose transport is activated hopping. That is a real class — doped Mott insulators, organic and molecular semiconductors, polaronic oxides, amorphous semiconductors — and this article set out to claim it. §4 measures the activation energy and finds 613–1723 meV, against the 10–500 meV those materials show. The claim does not survive its own measurement, and what the construction describes is the insulating end of the class and not the conducting one. Silicon and the III–Vs were never in scope, their gaps being band gaps.

3 The interface, and the one number that is wrong

The free interface has a defect of its own, recorded here because it conditions everything the partition does, though the hopping barrier does not turn on it. Partitioning a gas of density n gives a kinetic energy density $t(n) = C\, n^{5/3}$ — the Thomas–Fermi *form*, obtained from partition geometry rather than Fermi statistics — but the flat trial function is admissible on a free boundary, so $C = 0$ exactly. There is no degeneracy pressure, the screening length vanishes rather than being finite, and the plasmon does not disperse. The defect is one number and not the construction: C is fixed by the interface condition, and a Robin condition at $\beta a = 1.146$ gives $C = 2.871 = C_{\mathrm{TF}}$ exactly, against Dirichlet's 14.804, all three carrying the correct $n^{5/3}$ scaling.

Whether that value is tolerable for matter is the open question, and it is sharper than a parameter check. Adding the surface term and scanning it, H_2 unbinds by $\beta a \approx 0.71$, below the 1.146 the electron gas requires: Teller's theorem, that Thomas–Fermi binds no molecule, is met inside the framework rather than inherited from outside it. Double occupancy — the repair conduction would need at half filling — becomes favourable at $\beta a = 0.451$ and is affordable by comparison. These figures were obtained at differing densities and are not strictly commensurable; the equation of state is pursued elsewhere, and what matters here is only that the interface carries no cost of its own unless one is supplied.

A caution about specified partitions, learned by getting it wrong. The method of §4 evaluates the energy of a *stated* partition, and it is sound only when the stated partition is one the physics admits. The interface is not a surface to be shaped at will: the wavefunction is a single global field partitioned by labels, continuous across every interface, each domain carrying one unit of charge, and the boundary sits where the densities balance. Freezing a partition removes that last condition. An attempt to measure a bending rigidity by imposing a curvature, freezing it and relaxing the fields gave an energy falling symmetrically and quadratically in the imposed curvature, which reads convincingly as a negative rigidity; but released, a boundary started from the curved configuration returns to the flat one. The flat interface is the balance surface and is stable, and what the frozen scan measured was the energy gained by *dropping the balance condition.* A specified partition gives a legitimate energy difference between configurations the physics admits — as symmetry guarantees for the two in §4 — and a meaningless one along a direction it forbids. Nothing in the numbers distinguishes the two cases; the check is to release the boundary and see whether it stays.

4 Computing with it anyway

Electric conduction is the interaction of the electrons with the ion lattice. It is the elementary case — ordinary matter at ordinary temperature — and it is for that reason the *demanding* one here, since ordinary-temperature conduction is degenerate matter, and degeneracy is what this framework has least of.

One property of the construction is needed before anything else, and it is the property that makes conduction a live question rather than a hopeless one. With the free interface used throughout, a uniform density is admissible on any partition: $\psi_i \equiv |\Omega_i|^{-1/2}$ satisfies the normalisation, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Redrawing the boundary therefore costs nothing,

$$\text{a free interface between domains of equal density carries no confinement energy,} \tag{2}$$

and the boundary is the only object this framework moves — its solver evolves each interface by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, transferring ownership of a cell when the neighbour's density exceeds its own. Domains do not translate; they are redrawn. *A vanishing interface cost is a vanishing cost of transport.* What that same fact does to the equation of state is a separate matter and is taken up in §3, where it is much less welcome.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$. Within the scope set in §2 that is the *right* answer, activated transport being what a localised-carrier semiconductor does; it would be fatal only for a metal, which is why the target matters. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that reproduces alkali lattice constants is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near 2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $HF + H_2O \rightarrow F^- + H_3O^+$ and $HCl + NH_3 \rightarrow Cl^- + NH_4^+$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The cause is not what it first appeared. Kernel proximity assigns the labels only *initially*; thereafter a free-boundary rule evolves each interface from the local density balance and transfers ownership of a cell when the neighbouring domain's density exceeds its own. Partitions therefore do rearrange, and electron transfer is representable in principle. What defeated the attempts here was specific to each: a carrier given no kernel feels no force from the lattice; a bare kernel is given no domain, so a vacancy has nothing to migrate into; and a carrier placed on an occupied site can seize the nuclear region outright, since with free interfaces no confinement cost opposes it. Each is a property of the construction, not of (1).

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

Which filling the measurements belong to, and which needs the repair. It is worth being exact about this, because it determines how much of the article depends on β. At *half filling* — one electron per site, every domain occupied — the $U \to \infty$ construction forbids conduction outright: there is no vacancy for a carrier to enter, and the insulator is structural rather than dynamical. That is the case the double-occupancy repair is for. But it is not the case measured here. The chains reported above are doped by ionising every second atom, leaving half the sites empty, and every barrier in this article is the cost of a carrier moving into a *vacancy.* Single occupancy handles that without amendment.

So the transport results stand on their own: they require no finite U, no double occupancy and no β. What requires the repair is the undoped, half-filled chain, which is not run here. This is the stronger position of the two — the measurements are independent of the proposal made to extend them — and it locates precisely what the proposal would buy: not better hopping, but conduction at a filling where the construction currently permits none.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

What the barrier means as a field. A barrier becomes a threshold field when divided by the charge and the hop distance: $E_a/ea = 555\,\mathrm{meV}/6.0\,a_0 = 1.7 \times 10^9$ V/m, or 0.55 V per lattice site. Copper carrying an ampere runs at 4×10^{-12} V per site. The framework is therefore eleven orders of magnitude from conducting under any field a material survives, and the gap is not one a refinement closes.

Why a chain rather than a ring. A ring supplies a clean barrier — no ends, every site equivalent, the energy obliged to repeat with the lattice period — but it cannot host the drive: a conservative field has zero circulation, so no bias drives a sea around a closed loop. A chain has ends, a potential drop along it is an ordinary thing to apply, and a single carrier may enter at one end and leave at the other, which is the transport event itself.

The partition must be specified, not searched for. Before the number, the method, because the obvious approach fails and the failure is instructive. A kink is *not* the ground state, so a free-boundary minimisation relaxes it away — asked to hold one, the solver oscillates between the kinked and uniform partitions and the energy never settles, wandering over 0.05 Ha against a barrier of order 0.01. That is not a convergence failure to be tuned away: it is unaltered by the surface tension β, by reducing the boundary step fivefold, and by freezing the labels outright.

The cause is general and it disqualifies the free boundary for every transport quantity in this article. The interface moves by *local single-cell label flips*, so a run descends to whichever configuration no single flip improves — the nearest stuck point, not the minimum — and which one that is depends on the initial condition. Four independent geometries agree:

geometry	spread	against a barrier of
ring, between two builds	0.005 Ha	0.012–0.022
kink chain, across steps	0.05 Ha (never settles)	~ 0.01
$H^- + H$, four seeds	0.031 Ha	~ 0.001
alkali chain, two seeds	0.013 Ha	~ 0.001

In every case the ambiguity exceeds the quantity sought, by one to two orders. It is a failure of the *search*, not of the energy functional, which is why nothing in the functional touches it. It is also why the ring's 0.01186 Ha moved to 0.022 Ha under a recompilation that altered no physics: rounding at the last bit flips cells sitting near the threshold, and a different stuck point follows. The rigid-sliding figure should be read as a bound of that order rather than a determination, and the doping barriers of the earlier barrier scans, obtained the same way, with it.

The repair is not a better minimiser but no minimiser. The domain walls of a chain are planes perpendicular to the axis, so they can be *placed*: specify the partition, freeze it, and relax only the wavefunctions. Nothing is searched for, the same input returns the same output, and the question asked is no longer "where does the boundary go" but "what does this configuration cost" — which is the question a barrier actually poses. It is the device that makes the H_2 midplane exact, applied to a coordinate instead of a symmetry.

The hopping barrier, from two configurations neither of which was chosen. The quantity conduction depends on is the activation energy for a carrier to advance one site, and it can be had without choosing any partition, because *symmetry fixes both ends*. Take a chain with an odd number of sites carrying one vacancy, and let the carrier hop between the two central sites. Two configurations are then distinguished:

- the carrier centred **on** a site — invariant under reflection about that site;
- the carrier centred **between** the two sites — invariant under reflection about the bond midpoint.

Each is invariant under a reflection, hence a stationary point of the energy over partitions, hence a configuration at which the balance condition holds without being imposed. This is the footing that makes the H_2 midplane exact, and it is the only footing on which a specified partition is legitimate. The profile connecting the two is given the same width in both, so that position is the only thing that differs.

spacing a (a_0)	on site (Ha)	between sites (Ha)	barrier
4.0	−1.00761	−1.07092	0.06331 Ha = 1723 meV
5.0	−1.14835	−1.17695	0.02860 Ha = 778 meV
6.0	−1.21951	−1.24957	0.03006 Ha = 818 meV
7.0	−1.17275	−1.19527	0.02252 Ha = 613 meV

And the barrier must be minimised over the path, not read off one. A barrier is the lowest saddle connecting two states, not the highest point of whichever path one happens to sample. The family above has a second parameter — the width W over which the slip is spread — and fixing it describes a kink of rigid shape, which a minimum-energy path is under no obligation to be. Scanning it:

W (sites)	barrier	
0.4	929 meV	
0.8	818 meV	
1.5	**555 meV**	minimum; 552 at $n = 7$, 560 at $n = 11$
3.0	626 meV	

The curve has a minimum, which is what a Frenkel–Kontorova kink requires: elastic stiffness penalises a wide kink, the lattice's grip penalises a narrow one, and the physical width lies between. **The barrier is** $E_a \approx 555$ **meV** $= 21\, k_B T$, and that is the figure quoted hereafter — a minimax over the path family rather than a value at an arbitrary width, and independent of chain length to 1.5% at the width where it was checked twice.

The distinction matters for how the earlier numbers are read. Every figure obtained at a fixed W is an upper bound, since it is the cost along a rigid path; the 818 meV that this section reported before the width was varied is one such, too large by a third.

Three things follow, and the first is a structural claim rather than a number, though a narrower one than it first appeared. On a chain the carrier sits **between** sites: the bond-centred configuration is the minimum at every spacing tested, so the hop is bond to bond through the on-site saddle, which reproduces by an independent route the registry preference found above. That preference does *not* survive a second dimension. Repeating the same construction on a square lattice — identical carrier, identical slip, one dimension more — the hole sits on a *site* and the bond becomes the saddle, which is the conventional picture for vacancy hopping. The bond-centred result is a property of the chain and not of the construction, and it is recorded here as such.

Second, the barrier is a property of the hop and not of the chain. At $a = 6.0$ it is 0.03006 Ha for seven sites and 0.03247 Ha for nine — 8% apart, while the total energy moves by 0.59 Ha between them, a factor of 245 in sensitivity.

Third, and this is the substantive result, **compression raises the barrier**. It is 818 meV at the equilibrium spacing and 1723 meV compressed to $a = 4.0$, a doubling rather than a closing. That is the conclusion the compression scans of this article reached by the free-boundary route, and it is worth stating plainly that those scans are the ones §4 shows to be unreliable. The result survives the discrediting of the method that first found it.

Nor does doping, which was the last candidate. Carrier concentration was the one variable with evidence of moving the barrier downward: the free-boundary series of this article has 1.09 eV at 29% falling to 0.31 eV at 43%, a factor of 3.5. Repeating the measurement with symmetry-fixed endpoints — two vacancies in a seven-site chain, one hopping while the other stays — gives 1525 meV at 29% against 818 meV at 14%. The barrier *rises*, and the earlier trend is not reproduced. It belongs with the other figures obtained by that route.

One qualification belongs with the comparison. The carrier's preferred position changes with concentration: at 14% it sits at a bond and the site is the saddle, at 29% the reverse. The two hops are therefore not the same process, and 818 $\to$ 1525 should not be read as a clean trend. What matters for the conclusion is that both lie between one and one and a half electron-volts.

Two dimensions raise it further. The chain forces the carrier along one route, and one might expect a lattice to offer a cheaper one — an earlier scan of this framework found an *excess electron* threading between two atoms to cost 5.46 eV against 31.5 eV through one, a factor of 5.8. A hole has no such option. Built identically on a square lattice, the barrier is 2072 meV at 3×3 and 2911 meV at 5×5; the smaller lattice has a single interior site and is dominated by its edges, which is also where the bond-centred preference of the chain survives and then fails. Taking the larger figure, two dimensions make the hop *harder* by a factor of 3.6, not easier: a missing electron at a site disturbs four neighbours rather than two, and there is no route that avoids them.

The conclusion is therefore robust rather than provisional. Four variables have been tried, each of which could have produced a hopping semiconductor:

variable	effect on the barrier
lattice spacing, compressed	doubles it, 818 $\to$ 1723 meV
a second dimension	triples it, 818 $\to$ 2911 meV
carrier concentration	doubles it, 818 $\to$ 1525 meV
core radius	helps modestly

The barrier remains between 0.56 and 2.9 eV throughout, that is 21 to 113 $k_B T$, against the 10–500 meV of the materials this article hoped to describe. It is not one measurement that places the construction outside that class but four attempts to bring it inside, none of which succeeds, and two of which make matters worse in the direction one would naively expect to help. *No accessible variation reduces the barrier to the range where hopping occurs.*

What the barrier says about scope. Minimised over the path family the activation energy is 555 meV, and every variation tried leaves it between 555 and 1723 meV, that is 21 to 67 $k_B T$. Hopping semiconductors — the organic and molecular solids, the polaronic oxides, the amorphous semiconductors named in §2 — have activation energies of 10 to 500 meV. This construction is an order of magnitude above them at every accessible density, and shows no sign of approaching them under compression, which makes it worse rather than better.

The honest reading narrows the scope claim of §2 rather than supporting it. What this framework describes is a **Mott insulator**: a carrier localised on a site, a gap set by on-site repulsion, and a barrier too large for the carrier to move at any temperature at which the lattice survives. The threshold field tells the same story — $E_a/ea = 2.7 \times 10^9$ V/m, or 0.85 V per site against copper's 4×10^{-12}. Semiconduction was the class this article hoped to belong to; the measurement places it outside.

Earlier figures, and why they are not quoted. A series of barrier scans was made before the difficulty above was understood, all of them by the free-boundary route: a compression test on a ring, a doping series at two concentrations, a lattice barrier between interstitial pockets of order 5 eV, and a localised interface defect costing 179 meV independently of chain length. Every one specified or evolved a partition that no symmetry fixes, so each is a bound of its stated order rather than a determination, and all are superseded by the barrier above. What survives from them is the direction of the density trend, which the symmetry-fixed measurement confirms.

What is actually transported. The carrier in this framework is not an electron. An electron here is not a particle but a unit of charge density on a domain, and what exists is the *partition* of space; dynamics is the evolution of that partition, and transport is the partition pattern travelling. What moves along the chain is the **free boundary** — a locus where two densities balance — and nothing material traverses it at all.

The quantitative form of this is the useful one. As a kink passes, each electron advances by exactly one site spacing and then stops, while the kink advances the entire length of the chain. The carrier's displacement and its constituents' displacement are decoupled, and their ratio is the chain length. This is the relation between a dislocation and the slip it produces: the dislocation crosses the crystal, no atom moves far. Charge is genuinely transported and real charge arrives at the far end; it is the *carrier* that is immaterial, in the sense that a wave is immaterial while the water is not.

That diagnosis also accounts for the failures recorded earlier in this study rather than excusing them. Each of the unsuccessful constructions imposed a carrier that had to *translate*, moving bodily through occupied space, and returned costs of tens of electron-volts or else collapsed. Those were the price of forbidding the only mechanism the framework has. The solver states as much directly: its free boundary evolves by the density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, flipping ownership cell by cell. Boundary motion is the only motion it performs. The corona result is the same statement seen from the favourable side — an added electron spread into a uniform shell at no cost, which is a boundary redistributing freely.

This raises the stake of the kink measurement. If transport works here it works *because* of the free boundary, which is the single feature separating this framework from the standard theory; and if the boundary cannot carry charge cheaply, the difficulty is not in a detail of the construction but in the framework's defining object.

A repair that is not available, and one that is. An obvious response to the barrier is to let valence electrons share a territory, which removes it by construction. The $n^{5/3}$ scaling rules that out: sharing destroys the extensivity the partition supplies, and a metal would lose its kinetic energy density by fifteen orders of magnitude.

What the framework then is. Non-overlapping sites, occupation 0, 1 or 2, an on-site repulsion U, and transfer between neighbours: that is the **Hubbard model**. Which locates RealQM exactly. As formulated — one unit of charge per domain, always — it is the Hubbard model in the limit $U \to \infty$, where double occupancy is forbidden outright and every site is singly occupied. That limit is a Mott insulator at every density, which is precisely what the earlier barrier scans found by direct computation, and it explains the density trend as well: no finite U can be beaten by a bandwidth that does not exist.

The repair is correspondingly minimal. Keep the partition, keep the geometry, keep the Coulomb energetics, and allow occupation to take the value two at a computed cost. The framework then sits where the standard treatment of strongly correlated matter sits — the Hubbard model is the working description of Mott insulators, of the cuprates, and of the materials for which band theory fails — with the difference that its U and its lattice are derived rather than supplied.

We do not carry this out here. What is established is where the obstruction lies: not in the interface condition, not in the geometry, but in the rule of one unit of charge per domain. That rule is also the source of one of the framework's results, charge quantization as a consequence of indivisibility rather than an assumption. The tension is worth stating plainly: *the postulate that explains why charge comes in units is the postulate that forbids a metal.* Matter has it both ways, quantizing the particles while leaving the amplitude on each site continuous, and a partition of real densities has no evident means to do the same.

5 The same chain, computed the ordinary way

Everything above is internal to the framework. It is worth putting the same system beside the standard treatment, on quantities both can produce, and the result is a check on the scope claim of §2 rather than a contest.

Kohn–Sham DFT, PBE/6-31G, all electrons, a straight chain of lithium atoms at the spacings used above, with the geometry held uniform so that the comparison is like for like — a Peierls distortion is not permitted on either side:

	RealQM	PBE/6-31G	experiment
equilibrium spacing (a_0)	4.5	6.0	5.7 (bcc n.n.)
verdict at $a = 6.0$	pinned, 184 meV	metallic	lithium conducts
verdict at its own equilibrium	depinned, ≈ 0	metallic	—
cohesive energy (eV/atom)	~ 1.7	0.486	1.63 (bulk, 8 n.n.)

And it is not one functional's answer. Density-functional theory is not the Schrödinger equation: it replaces the interacting problem with fictitious non-interacting electrons in an effective potential, exact in principle by Hohenberg–Kohn but resting in practice on a *constructed* exchange–correlation functional and a finite basis — two uncontrolled approximations. The comparison above is therefore model against model, not model against exact, and would normally be qualified accordingly. On the quantity at issue it need not be. Repeating the scan with Hartree–Fock, which carries no correlation whatever, and with MP2, which carries it perturbatively, moves the total energy by some 0.3 Ha between methods and leaves the equilibrium spacing at 6.0 a_0 in all three:

a (a_0)	HF	PBE	MP2
4.5	−51.97126	−52.29669	−52.02861
5.0	−52.01844	−52.33386	−52.06750
6.0	**−52.04726**	**−52.34332**	**−52.08273**
7.0	−52.04089	−52.31653	−52.06476

The twenty per cent is thus a disagreement with standard quantum chemistry rather than with a choice of functional.

But the objection cuts the other way, and harder. If one may choose the approximation to suit the problem then one will always win, and that is not a comparison. Applied evenly, the charge falls on *this* side of the table. The standard column used a non-empirical functional, a general-purpose basis, and three treatments of correlation that were not selected after the fact — but the RealQM column carries a free parameter, the pseudopotential radius r_c, and it was set to 1.6 a_0 by choice rather than by any independent criterion. At $r_c = 2.4$ the equilibrium spacing of the same chain moves out beyond 6.0 a_0, which *agrees* with the standard result. The twenty per cent is therefore a statement about $r_c = 1.6$ and not about the framework, and the framework could have been made to agree as easily as to disagree.

A comparison is only worth making if the parameters are fixed by something other than the quantity being compared. The proper protocol is to calibrate r_c on the *isolated* lithium atom — requiring the single pseudo-atom to reproduce the measured first ionisation energy, 0.198 Ha — and then to compute the chain with that value and no further adjustment. The chain is then a prediction rather than a fit, and stands on the same footing as the standard column, in which nothing was tuned to lithium chains. Until that is done the table above should be read as establishing the *method-independence of the standard answer* and nothing about the size of any discrepancy.
The metallic verdict is not read off a single gap, which any finite chain has, but off its scaling: 0.432, 0.346, 0.291, 0.247, 0.173 eV at $n = 5, 7, 9, 11, 13$, with $n \times$ gap constant near 2.4 eV, so the gap closes as $1/n$ and is finite-size rather than a band gap.

That verdict is weaker than it looks, and in a direction that matters. What was compared are Kohn–Sham eigenvalue differences, and Kohn–Sham eigenvalues are not excitation energies. They belong to a fictitious system of *non-interacting* particles introduced to reproduce a density; the true gap differs from theirs by the derivative discontinuity of the exchange–correlation functional, which PBE does not possess at all. Approximate functionals are known to underestimate gaps for exactly this reason, often by a factor of two. A small Kohn–Sham gap is therefore what the method returns whether or not a gap is present.

The bias has a direction, and it is the one at issue here. Approximate functionals carry a self-interaction error — an electron repelled in part by its own density — whose consequence is **delocalisation error**: densities spread too far, localised solutions are penalised, and metallic answers are favoured. So the standard column calling this chain metallic is that treatment erring in its characteristic direction, not a neutral measurement. Against it, the framework of this article over-localises, by construction, being $U \to \infty$. *Both are biased and in opposite senses*, and lithium is simply a case where the standard bias does no damage, because the chain really is metallic. On a system where it does damage — an oxide, a doped Mott insulator, anything whose gap is on-site repulsion — the delocalisation error is the well-known failure and the localisation is the correct instinct.

Two further asymmetries belong in the record. Density-functional theory admits no convergence path: wavefunction methods form a hierarchy, Hartree–Fock through MP2, CCSD, CCSD(T) to full configuration interaction, which converges to the exact answer within a basis, whereas no procedure refines PBE towards the true functional. And the quantities are not of the same kind. What is computed here are total energies of *specified configurations* — differences between states of one system, which are observables — where the gap above is an eigenvalue difference in an auxiliary system that was never claimed to have physical states. Where this framework is less accurate it is at least computing something that means what it says.

What this establishes, and what it does not. On this system the standard treatment is better and cheaply so: the equilibrium spacing within a few per cent of the measured nearest-neighbour distance, the right conduction verdict, a sensible cohesive energy, in seconds. RealQM places the atoms some twenty per cent too close, and at the spacing lithium actually adopts it predicts a pinning barrier of 184 meV where the metal has none.

That is not a surprise and should not be read as one, because *lithium is outside the scope this article claims*. §2 excludes metals in kind and not by degree: extended Bloch states, band transport, a Fermi surface, none of which the construction possesses. A simple metal is the standard treatment's best case and this framework's declared worst, so the table is a scope check and not a competition.

As a scope check it passes, in the specific sense that matters: the framework errs by *over-localising*, in bond length and in pinning alike, which is what $U \to \infty$ implies and therefore the direction the identification predicts. It fails as it was said it would, rather than arbitrarily.

Where the comparison would actually bite. The corollary is uncomfortable and is recorded plainly: nothing computed here is a quantity the standard treatment does not already supply more accurately. The case for the construction rests instead on the real-space mechanism — a registry, a sliding partition, a carrier that is a boundary rather than a particle — being a more transparent account of activated hopping than a model Hamiltonian with fitted parameters. That claim is not tested by lithium. It would be tested on the systems the scope table names, where on-site repulsion rather than band structure sets the gap and where density-functional theory is itself in difficulty: a doped Mott insulator, an organic semiconductor, a polaronic oxide. Those comparisons are not made here.

A note on what was tested numerically

The plasma frequency $\omega_p^2 = 4\pi n$, the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. The alkali lattice constants are a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Three attempts are reported as null rather than omitted. The interface parameter β was tested on helium at three meshes at fixed imaginary time, giving shifts of $+0.151$, -0.018 and -0.184 Ha as the mesh refined — swinging through zero and growing, with absolute energies moving away from the exact -2.9037. The cause is the geometry rather than the arithmetic: in helium the two domains meet at a plane *through the nucleus*, so interface error and Coulomb-cusp error are superimposed and cannot be separated on a uniform grid. The measurement wants H_2, where the domains meet at the midplane *between* the nuclei in a smooth region. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the constructions rather than by the runs: a carrier without a kernel feels no lattice force, and a bare kernel receives no domain for a vacancy to occupy. Both are recorded because the null results located those obstructions. A caution on the numerics belongs with them. Several attempts returned energies below the physical floor for their particle content, but these track the boundary speed and the seeding rather than the configuration: with the interface driven faster than the densities can follow, or with a carrier seeded directly onto an occupied kernel, the relaxation runs away, while the same configurations at a quarter of that speed settled to physical values. A two-domain shell on one nucleus is in fact stable under the density-balance rule — compressing the inner domain raises its density at the interface and lowers the outer's, which pushes the boundary back out — and both domains carry real kinetic energy there, since the nucleus shapes their profiles. The vanishing of C is a statement about a *uniform* density and does not transfer to a shaped one. Scripts and pages accompany the source.

6 Conclusion

The question was whether a construction of non-overlapping unit charge densities, meeting at free boundaries, conducts. The answer is a number: the activation energy for its carrier to advance one site is $E_a = 555$ meV $= 21\,k_BT$, obtained from two configurations that symmetry fixes and minimised over the width of the path connecting them, independent of chain length to 1.5%. That is above the 10–500 meV of the hopping semiconductors this article set out to describe, though not by the order of magnitude a single fixed-width calculation would have suggested, and no variation tried — compression, a second dimension, carrier concentration, core radius — reduces it. *The construction does not conduct at any temperature at which a lattice survives.*

Three things stand behind that number, and they are of different kinds.

The carrier is a free boundary, and that picture holds. An electron here is a unit of charge density on a domain; what exists is the partition of space. Transport is therefore the partition pattern travelling, in the relation a dislocation bears to slip: charge arrives at the far end while nothing material crosses the sample. That diagnosis accounts for the null results reported above, each of which had imposed a carrier obliged to translate bodily through occupied space, and it was confirmed directly — given a geometry that permits it, the boundary migrates and the charge goes with it. The picture is not in question.

Costing it defeats a minimisation, and that is the article's principal result. A free boundary evolves by local rules, so a run descends to the nearest configuration no single cell change improves. That is a stuck point and not a minimum, and which one is reached depends on the initial condition. Measured across four independent geometries the ambiguity is 0.005–0.05 Ha against barriers of 0.001–0.02: one to two orders larger than the quantity being sought, unrepaired by the surface tension, by the boundary step, by freezing the labels, or by changing to a system without the first one's pathologies. It is a failure of the search and not of the energy functional, which is why nothing in the functional touches it, and it invalidates transport energies obtained this way — including, by its own logic, the rigid-sliding figure of this article, which moved under a recompilation that altered no physics.

The repair is not a better minimiser but none. Domain walls are surfaces and can be placed: specify the partition, freeze it, relax only the wavefunctions, and ask what a stated configuration costs rather than where a boundary goes. Repeated runs then agree to every decimal recorded. What that yields is modest but real — a misplaced domain wall at 179 meV, independent of chain length to 5%; a vacancy's mismatch gaining 0.112 Ha by delocalising, also length-independent; a sliding barrier tunable through zero by density and by core radius. What it does not yet yield is the activation energy for a kink to *move*, which is the quantity conductivity depends on, and which is left open here with a construction stated rather than a number estimated.

What the postulate buys, and what it costs. The results here say something about non-overlap itself, and it is worth stating separately from any of them. Overlap is not an optional feature of quantum mechanics; it is the mechanism by which amplitude passes between places. Forbidding it makes the electrons objects with disjoint supports — distinguishable, in the way classical charges are — and leaves a theory of *static* charge distributions carrying a quantum-derived kinetic penalty.

That division predicts where the framework succeeds and where it does not, and the prediction is borne out across its whole corpus. Ground-state energies, bond lengths, lattice constants, binding curves: quantities set by electrostatics against a localisation cost, and it does them well. Tunnelling, exchange, dispersion, transport, a Fermi surface: quantities that exist only because amplitude is shared, and it has none of them. The failure recorded in this article is not a shortfall in accuracy but an absence of the ingredient, and it could have been predicted from the postulate without any computation — which is the strongest form of the result, and the least comfortable.

The construction is $U \to \infty$, and this sets both its scope and its errors. Single occupancy is the Hubbard limit in which double occupancy is forbidden, so metals are excluded in kind — no extended states, no band transport, no Fermi surface — and what remains in scope is matter whose gap is on-site repulsion. The same fact fixes the direction of every discrepancy measured against standard treatments: the framework *over-localises*, in bond length and in pinning alike. It fails as its own analysis says it should, which is worth more than an accidental agreement, but it does not yet produce a quantity that established methods do not supply more accurately. Its case rests on the mechanism being a clearer account of activated hopping than a model Hamiltonian with fitted parameters — a claim tested not by simple metals, where density-functional theory is at its best, but by the systems where on-site repulsion sets the gap and that theory is itself in difficulty.

Two repairs were identified and priced. Finite U in place of infinite, which conduction at half filling requires, becomes favourable at $\beta a = 0.451$ and leaves molecules bound. Restoring the interface coefficient to its Thomas–Fermi value requires $\beta a = 1.146$, and H_2 unbinds by $\beta a \approx 0.71$. Teller's theorem is thereby met inside the framework rather than inherited from outside it.

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